

Questions: What do you mean by Information System and Technology?

Answer:

An **Information System** means a group of things (like people, computers, and processes) that work together to collect, store, process, and share information.

The main goal of an information system is to help organizations make decisions, solve problems, and manage operations more efficiently.

Information Technology (IT) is the part of an information system that deals only with the technology — like computers, mobile phones, software, internet, and networks. It is the "tools" that help to handle the information.

Example:

Suppose a hospital uses a computer system to record patients' details, doctors' schedules, and billing.

- The **Information System** includes doctors, nurses, software, computers, and the rules they follow to manage patients.
- The **Information Technology** includes the computers, hospital management software, and internet connections.

Questions: A dollar today is worth more than a dollar one year from now.

Answer:

Money has a time value, meaning that a dollar today is worth more than a dollar in the future. This is because money received today can be invested to earn interest or generate returns, whereas future money cannot be used immediately. Therefore, when we expect to receive a certain amount in the future, we calculate its present value (PV) to know what it is worth today. The formula to find present value is:

We know that

$$PV = \frac{FV}{(1 + r)^n}$$

Where:

✓ PV = Present value

✓ FV = Future value (the amount of money you want to calculate the present value for)

✓ r = Interest rate or discount rate (expressed as a decimal)

✓ n = Number of periods (usually in years)

$$PV = \frac{\$1000}{(1 + 0.05)^3}$$
$$PV \approx \$863.84$$

Thus, the present value of \$1,000 to be received after three years, with a 5% discount rate, is approximately **\$863.84**. This means that \$863.84 today is equivalent to \$1,000 received three years later under the given conditions.

Question: How many symbols are used for DFD? Make comparison between DFDs and ERD?

Answer:

Three symbols and one connection:

1.External entities are represented by squares as the source or destination of data



(External Entity)

2.Processes are represented by rectangles with rounded corners.



(Process)

3.Data Flows are referred to by arrows to denote the physical or electronic flow of



(Data Flow)

data.

4.Data Stores are physical or electronic-like XML files denoted by open-ended rectangles.



(Data Store)

DFD

It stands for **Data Flow Diagram**.

Main objective is to represent the **processes** and **data flow** between them.

It explains the flow and process of **data input**, **data output**, and **storing data**.

Rule: Every **process** should have at least one **input** and one **output**.

Used to design the **data flow** of a system.

ERD

It stands for **Entity Relationship Diagram** or **Model**.

Main objective is to represent the **data objects (entities)** and their **relationships**.

It explains and represents the **relationship** between **entities** stored in a **database**.

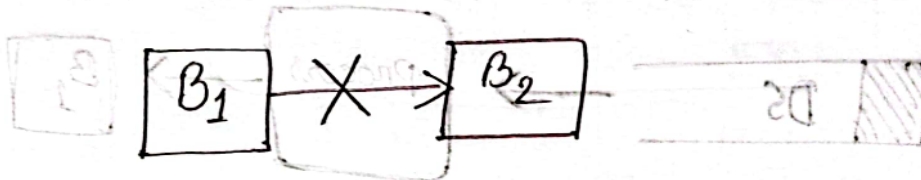
Rule: Each **entity** must be a **group of similar things**.

Used to design the **database structure**.

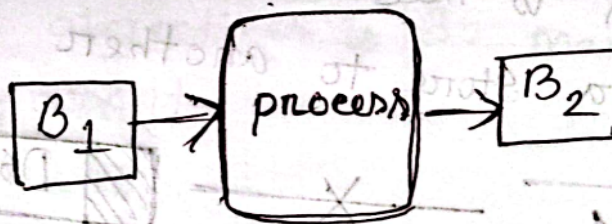
⑪ What are the cases where illegal data flows happen?

Data flow analysis is that all flows must begin with or end at a processing step.

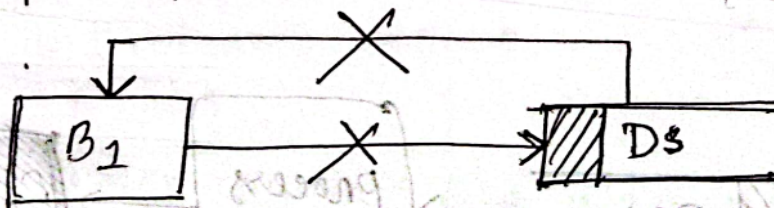
① A process is needed to exchange data flows between external agents.



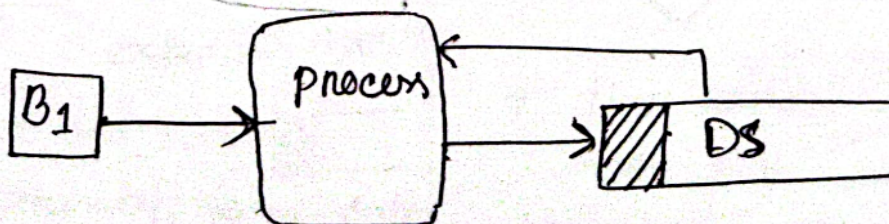
Correct data flows:



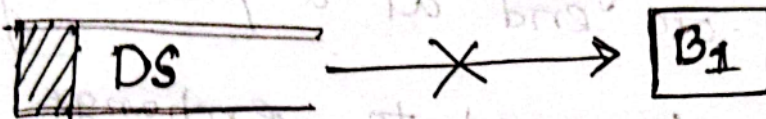
② A process is needed to update (or use) a data store.



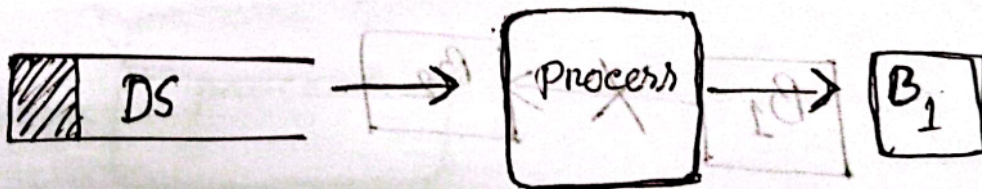
Correct data flows:



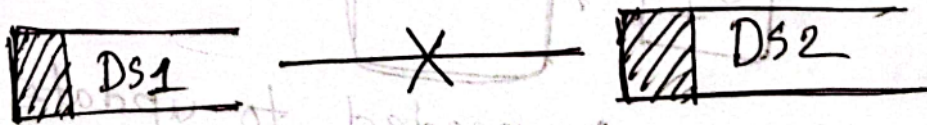
- ③ A process is needed to present data from a data store.



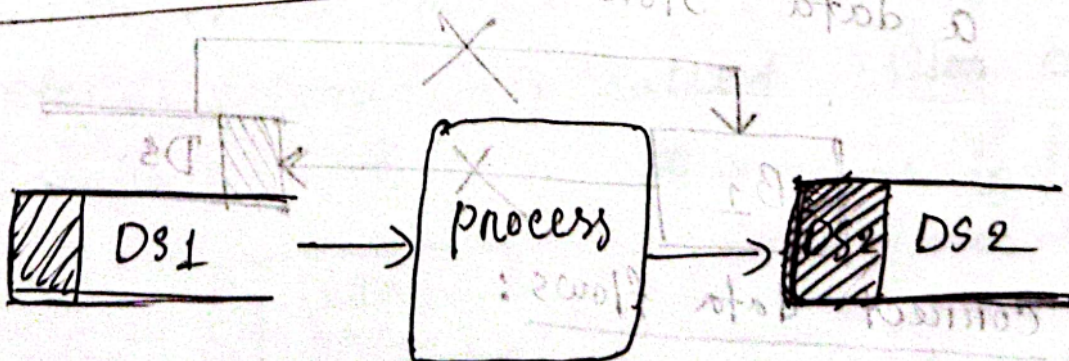
correct data flows:



- ④ A process is needed to move data from one data store to another



correct data flows:



- **No direct flow** between data stores.
- **No direct flow** between external entities.
- **No direct flow** from data store to external entity.
- **Each process** must have at least **one input and one output**.

Question: Describe different types of Questionnaire?

Answer: A **questionnaire** is a set of written questions used to **collect information** from people.

It is often used in **research, surveys, and information systems** to understand people's opinions, experiences, or needs.

There are two types:

1. Free Format
2. Fixed format

Free Format:

They give the person more freedom to answer.

A question is asked, and the person writes the answer in the blank space given after the question.

Question: What do you like to do in your free time?

Answer: I like to read books and play football with my friends.

Types of free-format questions:

- i. Opinion question.
- ii. Exploratory question.
- iii. Scenario-based question.
- iv. Descriptive questions.
- v. Comparative questions.

Fixed format:

Fixed format questionnaires give the person choices to select from.

The person cannot write their own answer.

They must choose one answer from the given options.

This type is easy to answer and easy to check or count.

Type of fixed format questions:

- i) Multiple choice Questions
- ii) Rating Questions
- iii) Ranking Questions

Question: What are the advantage and disadvantage of Questionnaire?

Advantages:

1. Quick to get answers from many people.
2. Low cost, as you don't need to spend much money.
3. Easy to compare answers from everyone.
4. Anonymous responses, so people can be honest.
5. Can be sent to many people, like online, by email, or on paper.

Disadvantages:

1. Limited answers (e.g., in multiple-choice, there are only a few options).
2. Low response rate, meaning not everyone may answer.
3. People may misunderstand the questions.
4. Some answers may not be detailed enough.
5. There is no one to help if someone doesn't understand the question.

Question: What are the advantages and disadvantages Of interviews?

Advantage:

- i. **Motivates the Interviewee:** Interviews help encourage the person to share more.
- ii. **Detailed Information:** You can get more detailed answers than other methods.
- iii. **Flexible Questions:** Questions can be changed based on the answers.
- iv. **Body Language:** You can see the person's body language to understand better.

Disadvantage:

- i. **Takes Time:** Interviews use a lot of time.
- ii. **Depends on Skills:** The success of an interview depends on how good the interviewer is.
- iii. **Location Problems:** Interviews can be hard if the person is far away.

⑦ What should do and avoid for interview session?

Do	Don't (Avoid)
1. Be polite	1. continuing an interview unnecessarily
2. Listen carefully	2. Assuming an answer is finished.
3. Maintain control	3. Revealing revealing verbal and nonverbal clues.
4. Inquiry	4. Revealing your personal biases.
5. observe mannerisms and nonverbal communication	5. Talking instead of listening.
6. Be patient	6. Assuming anything about the topic and the interviewee.
7. Keep interviewee at comfort	7. Tap recording - a sign of poor listening skills.
8. Maintain self control.	

Brainstorming

Brainstorming is a group activity where people share many ideas quickly to solve a problem. No one judges the ideas until all are written down.

Brainstorming Guidelines (Step-by-step):

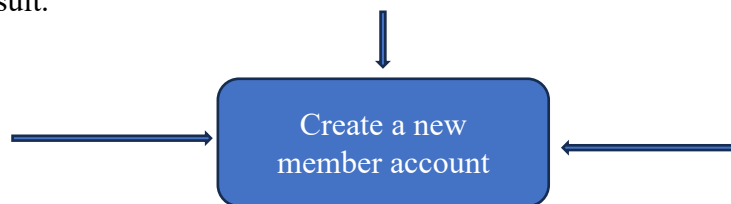
1. Sit together in a quiet place.
2. Make sure everyone knows the topic.
3. Choose one person to write down the ideas.
4. Remind everyone of the brainstorming rules (no judging ideas).
5. In a short time, all members say their ideas quickly.
6. After all ideas are written, then discuss and judge them.
7. Mix and improve the good ideas.

Common Process Errors in Data Flow Diagrams (DFDs)

1. Black Hole:

A process has inputs but no outputs.

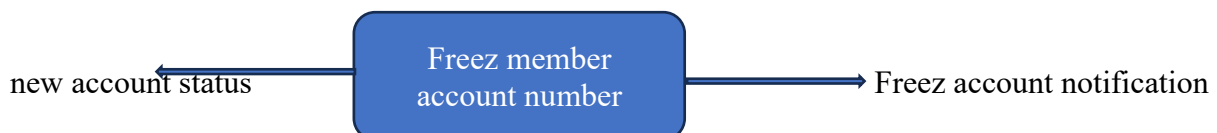
👉 It is called a *black hole* because data enters the process but disappears without any result.



2. David Copperfield (Miracle):

A process has outputs but no inputs.

👉 Unless you are David Copperfield (most commercially successful magician in history) it's a miracle.



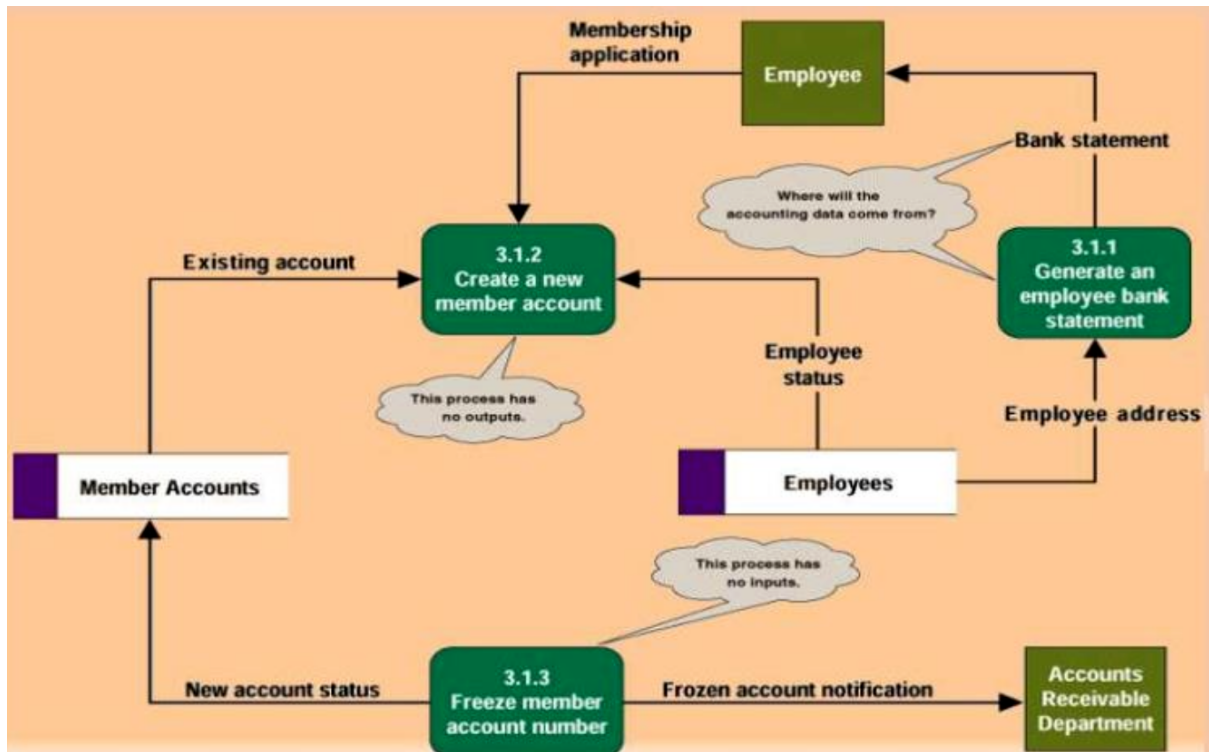
3. Gray Hole:

A process has inputs, but they are **not enough** to produce the outputs.

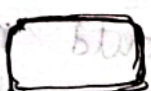
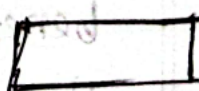
👉 The result does not match the input due to:

- Wrong process name

- Wrong or missing inputs/outputs
- Incomplete or illogical information



⑫ ~~Make~~ Make Comparison between DFD and ERD

DFD	ERD
1. It stands for Data Flow Diagram.	1. It stands for Entity Relationship Diagram.
2. DFD shows the flow of Data through a system.	2. ERD shows the relationship between entities.
3. main objectives is to represent the <u>process and data flow</u> between them.	3. main objective is to represent <u>data object</u> .
4. Symbol of DFD: * round rectangle  represent process * square represent  external Agents * open end box  represent store data * Arrow → represent flow of data	4. Symbol of ERD: * rectangles  represent entity. * diamond box represent  relationship * line and standard notation represent cardinality.

A dollar today is worth more than a dollar one year from now", including the **significance** and **example**.

Significance of the Statement:

"A dollar today is worth more than a dollar in the future" because:

1. **Opportunity Cost:**
You can **invest the money today** and earn interest or profit.
Example: \$1 today can become \$1.10 in a year if the interest rate is 10%.
 2. **Time Value of Money:**
Money loses value over time because of changing interest rates and future uncertainty.
 3. **Inflation:**
Prices increase over time.
The same \$1 today will buy **more goods** than the same \$1 next year.
-

Example 1:

Suppose you will receive **\$20,000** after **2 years**, and the interest rate is **10% per year**.
Let's find out how much money you'd need **today** to have \$20,000 in 2 years.

Formula:

$$\text{Present Value (PV)} = \frac{20000}{(1 + 0.10)^2} = \frac{20000}{1.21} \approx 16528$$

So, if you invest **\$16,528 today** at 10% interest, you will get **\$20,000 in 2 years**.

Example 2:

You can choose:

- **\$1000 today**, or
- **\$1000 one year later**

Better to take **\$1000 today** because you can invest it, use it, or it will have more value now.

Give an outline for standard oral presentation.

Oral Presentation Structure:

Introduction (1/6 of total time available)

- A. Problem Statement: কী সমস্যা রয়েছে বা কেন এই প্রজেক্ট দরকার?
- B. Work Completed to Date: এখন পর্যন্ত কী কী কাজ শেষ করেছো?

Main Part of the Presentation (2/3 of total time available)

- A. Existing Problems and Limitations: আগের সিস্টেম বা বর্তমান অবস্থা কীভাবে খারাপ বা সীমাবদ্ধ?
- B. Proposed System: তুমি যে নতুন সিস্টেম প্রস্তাব করছো সেটা কেমন হবে?
- C. Feasibility Analysis: এটা বানানো কতটা সম্ভব, লাভজনক এবং বাস্তবসম্মত?
- D. Proposed Schedule: ভবিষ্যতে কাজগুলো কবে কবে শেষ করবে? সময়সূচি দাও।

Questions and Concerns from the Audience (Not fixed time)

- * উপস্থাপনার সময়ের মধ্যে ধরতে হবে না।
- * দর্শক বা শ্রোতা তোমাকে প্রশ্ন করবে বা তাদের উদ্বেগ জানাবে।
- * তুমি উত্তর দিবে।

Conclusion (1/6 of total time available)

- A. Summary of Proposal: তুমি কী কী বললে তার একটা ছোট সারসংক্ষেপ।
- B. Call to Action: এখন তুমি চাইবে কর্তৃপক্ষ যেন তোমাকে অনুমতি দেয় বা সহযোগিতা করে প্রজেক্ট চালিয়ে যেতে।

Comparison between Factual and Administrative Report Format

Factual Format	Administrative Format
1. Starts with Introduction	1. Starts with <u>Introduction</u>
2. Shows Methods and Procedures	2. Gives Conclusions and Recommendations early
3. Presents Facts and Details	3. Shows Summary and Discussion of facts
4. <u>Discusses and analyzes the facts</u>	4. Describes Methods and Procedures later
5. Gives Recommendations	5. Ends with <u>Final Conclusion</u>
6. Ends with Conclusion	6. Adds Appendices with <u>facts and extra info</u>

Main Difference in One Line:

Factual format provides facts first, followed by recommendations, while **Administrative format** starts with conclusions for quick review.

difference between **databases** and **conventional files**:

D D SM D BRB

Topic	Databases	Conventional Files
Data Storage	Stores data in tables (like Excel)	Stores data in plain files (text or binary)
Data Repeating	Less repeating (no duplicate data)	More repeating (same data again and again)
Speed	Fast to search and update data	Slower to search and update
Many Users	Many people can use at the same time	Hard to use by many at once
Data Safety	More secure with login and rules	Less secure
Backup	Easy to take backups	Needs manual work
Relationships	Can connect different tables easily	Cannot connect files directly
Best For	Big systems like banks, schools, hospitals	Small tasks like notes or log files

In short:

- A **database** is smart and good for big, organized data.
- A **conventional file** is simple and good for small, basic tasks.

What is Present Value (PV)?

Present Value means:

Today's value of an amount of money that you will receive in the **future**.

Because of **interest** or **inflation**, money today is more valuable than money in the future.

Formula:

$$PV = \frac{FV}{(1 + r)^n}$$

Where:

- **PV** = Present Value (today's value)
- **FV** = Future Value (money you'll get later)
- **r** = Interest or discount rate (in decimal, e.g., 20% = 0.20)
- **n** = Number of years

Example:

Suppose:

- Future Value (FV) = ₦10,000
- Rate (r) = 20% = 0.20
- Time (n) = 5 years

We calculate:

$$PV = 10000(1+0.2)^{-5} = 10000 \times 0.68059 \approx 6805.9$$
$$PV = \frac{10000}{(1 + 0.2)^5} = \frac{10000}{2.48832} \approx 4017.19$$

So, **₹10,000 after 5 years** is worth about **₹4017.19 today**.

(৫ বছর পর ৳১০,০০০ এর মূল্য আজ প্রায় ৳৪০১৭.১৯)

How can we calculate the Present Value of money?

Present Value (PV) is the current value of a future amount of money, calculated using a discount rate. It shows how much future money is worth today.

Formula:

$$PV = \frac{FV}{(1 + r)^n}$$

Where:

- **PV** = Present Value
- **FV** = Future Value
- **i** = Discount rate
- **n** = Number of years

Steps:

1. Write the Future Value (FV).
2. Use the discount rate (i).
3. Use the number of periods (n).
4. Apply the formula to get PV.

Example:

If $FV = \$1$, $i = 12\% = 0.12$, and $n = 2$,

$$PV = 1(1+0.12)^{-2} = 1 \times 0.797 \approx 0.797$$
$$PV = \frac{1}{(1 + 0.12)^2} = \frac{1}{1.2544} \approx 0.797$$

So, the present value of \$1 after 2 years is **\$0.797**.

The time value of money is not taken into account for payback analysis - Explain the statement with appropriate example.

The Payback Period is a simple method used to determine when an investment will pay back its initial cost.

It calculates the amount of time it takes to recover the original investment through future cash inflows.

significant drawback with the Payback Period Method:

1. **It doesn't consider the time value of money:** Payback Period assumes that money received in the future is just as valuable as money received today. But in reality, money today is worth more because you can earn interest or returns on it.
2. **It doesn't adjust future cash inflows:** The method treats future cash inflows the same as current ones, which is not correct.

Time Value of Money (TVM)

The Time Value of Money states that money today is worth more than the same amount of money in the future because it has the potential to earn interest or returns over time.

Example:

Consider an investment of \$1,000, which provides an annual cash flow of \$500.

- **Initial Investment:** \$1,000
- **Annual Cash Flow:** \$500

Payback Period = $1000 / 500 = 2$ years

So, the investment pays back in 2 years.

Problem with Payback Period:

The Payback Period method does not consider the time value of money. Money today (like the \$500 you get in the first year) is worth more than the same amount of money in the future (like the \$500 you get after 2 years).

The method assumes that cash flows are equal in value, whether they happen today or in the future, but this is not correct.